DP BIOLOGY | SKILLS 2 OF 3

Tools for the lab and field

How do we measure what we cannot see?
Which tool suits which question?

Each tool links to the subtopics where it is used. All examples use invented data.



Author not stated, Public domain

What you will learn



Author not stated, Public domain

1 Microscopes

Prepare slides, calibrate a graticule, and state uncertainty



Jahra Tasfia Reza, CC0

2 Solutions

Make dilutions, and calculate Rf from a chromatogram



U.S. Fish and Wildlife Service – Midwe, Public domain

3 Instruments

Colorimeters, respirometers, loggers, PCR and gels



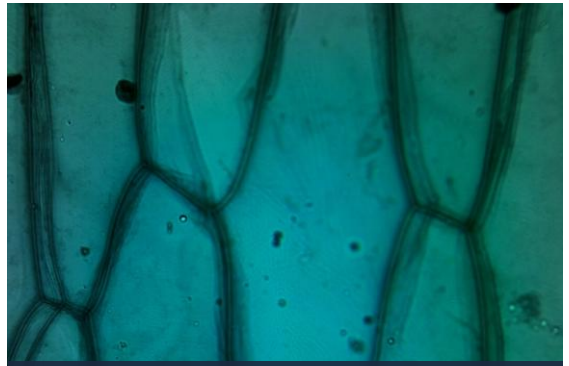
YellowstoneNPS, Public domain

4 Fieldwork

Choose a field method, and the right graph

Contents

In PowerPoint, click a picture to jump to that section. On the website, use Next and Back.



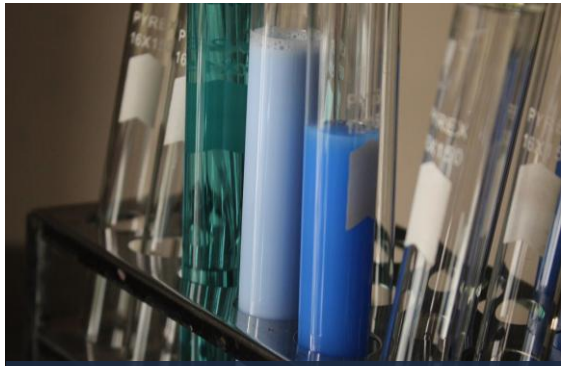
Light microscopy



Calibrating a graticule



Measuring



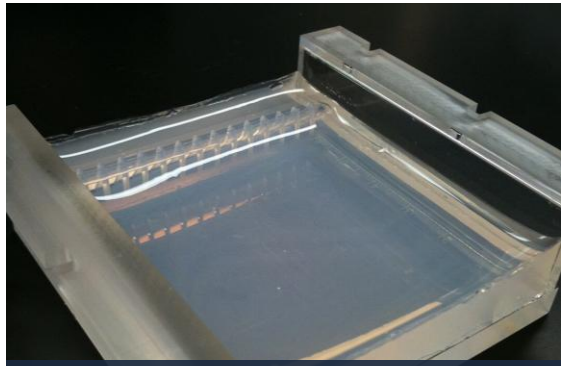
Dilutions



Chromatography



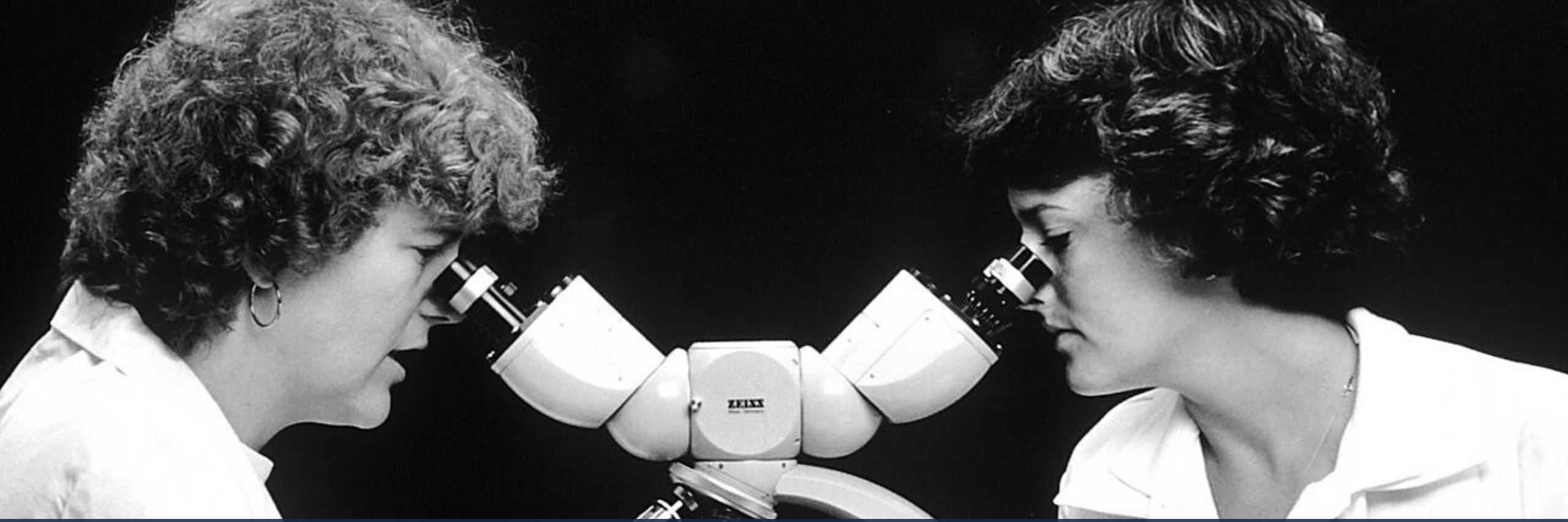
Colorimetry and loggers



PCR and gels



Fieldwork and graphs



01

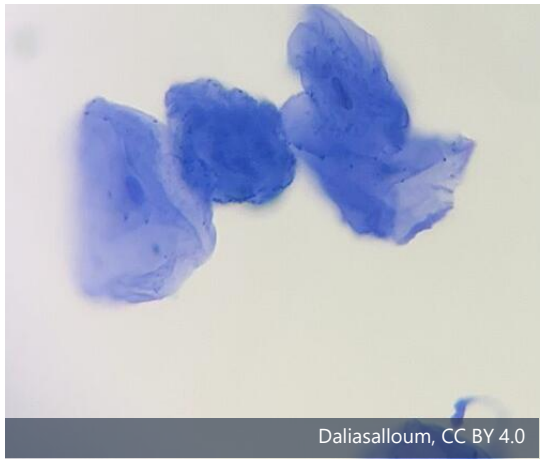
Microscopes and measuring

Cells are too small to see, so we magnify them and measure them carefully.

Author not stated, Public domain

Light microscopy

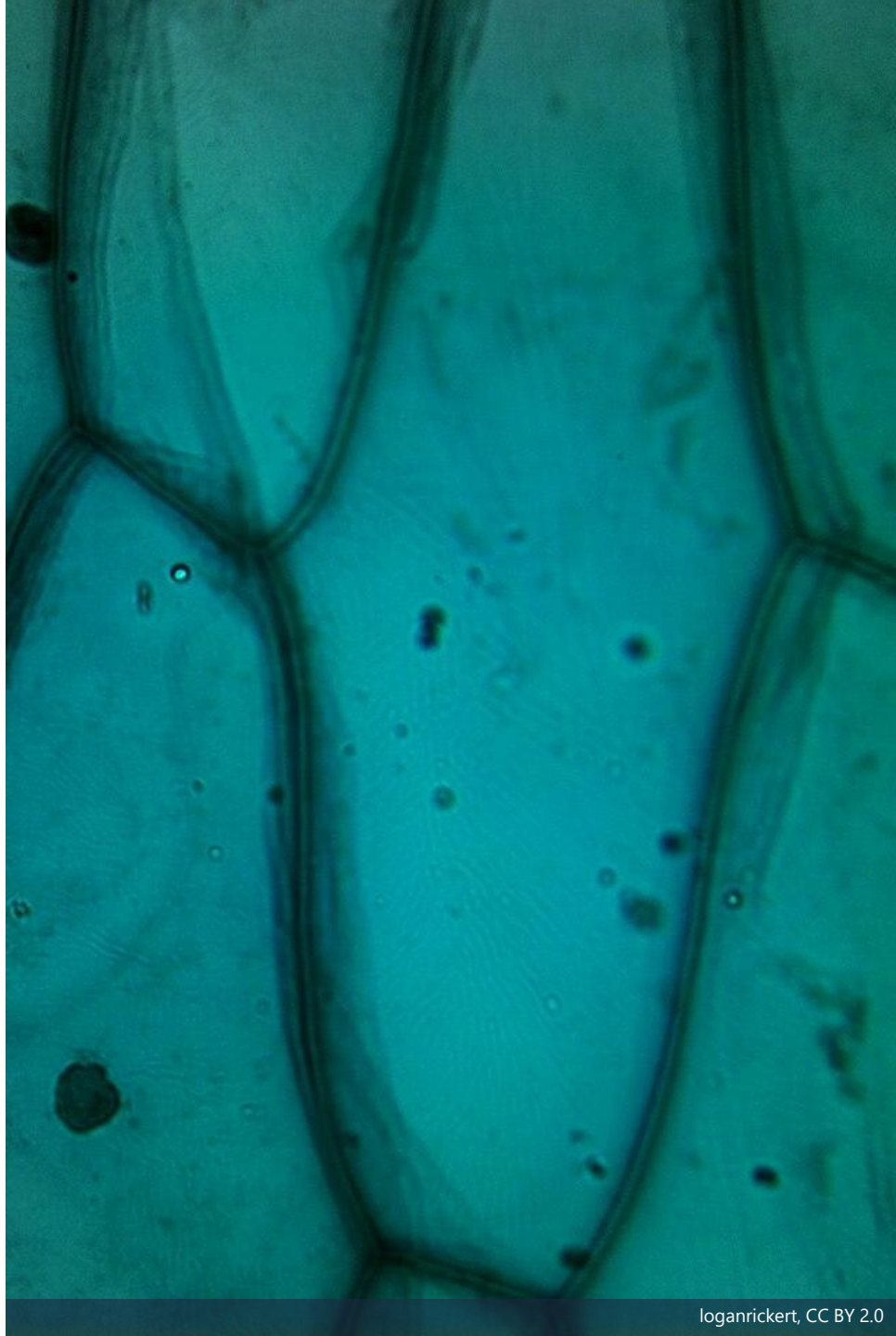
1. Cut a **thin** layer
2. Add a **stain**
3. Lower the coverslip at an angle
4. Low power first, then **fine focus**



Daliasalloum, CC BY 4.0

Stains

Iodine shows starch grains in plant cells. Methylene blue shows nuclei in animal cells, such as the cheek cells on the left. Draw what you see, with a scale bar. Used in A2.2 Cell structure.



loganrickert, CC BY 2.0

Calibrating an eyepiece graticule



RIT RAJARSHI, CC BY 4.0

Stage micrometer: 1 division = 10 µm



Eyepiece graticule: 25 units line up with 10 divisions



1 unit = 4 µm

10 divisions × 10 µm = 100 µm, shared by 25 eyepiece units.
100 / 25 = 4 µm per unit.

Remember

Recalibrate for every objective lens: the graticule stays the same size, but the specimen is magnified more. Then measure a cell in units and multiply: 6 units × 4 µm = 24 µm.

TRY IT

Calibrate and measure

With the x40 objective, 40 eyepiece units line up with 4 stage micrometer divisions of 10 µm each.

1. What is one eyepiece unit worth?
2. A cheek cell is 55 eyepiece units across. How wide is it?

> Hint: First find the total length of the micrometer divisions.

The answer is on the next slide.

Daliasalloum, CC BY 4.0

ANSWER

Calibrate and measure

1. 4 × 10 µm = 40 µm, so 1 eyepiece unit = 40 / 40 = 1 µm
2. 55 units × 1 µm = 55 µm

Compare your working with each step.

Key point

Cheek cells are typically 50 to 60 µm across, so the answer is sensible. Always check that a size is realistic.

Measuring and uncertainty



IAEA Imagebank, CC BY 2.0

Burette

Read the bottom of the meniscus at eye level. Scale every 0.1 cm^3: +/- 0.05 cm^3.



Diane A. Reid, Public domain

Pipette

Best for small, exact volumes. Choose one close to the volume you need.



QuodScripsiScripsi, CC0

Water bath

Thermometer every 1 °C: +/- 0.5 °C. Check the bath, not just the dial.



Stefan (Diskussion). Original uploader, Public domain

Stopwatch

Reads to 0.01 s, but human reaction time is much larger.

Convention

Analogue scale: +/- half the smallest division. Digital display: +/- the last digit. Record every value to the resolution of the instrument, with units in the column heading only.



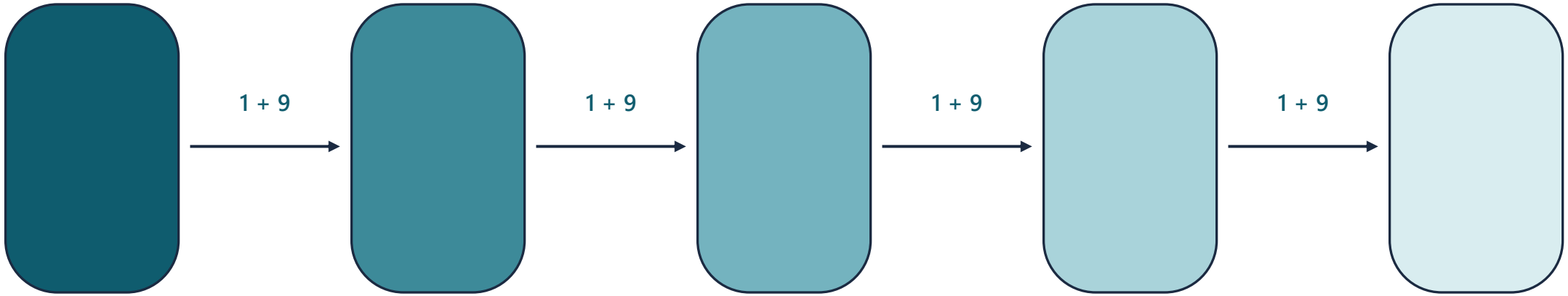
02

Solutions and separation

Make the concentrations you need, then pull mixtures apart.

Jahra Tasfia Reza, CC0

Dilutions



Serial dilution: each step, 1 cm³ of the last solution + 9 cm³ of water.

Target	Stock 1.0	Water
0.2 mol dm⁻³	2 cm³	8 cm³
0.6 mol dm⁻³	6 cm³	4 cm³

Linear series

For evenly spaced concentrations, mix stock and water in proportion to a fixed total. Used in D2.3 Water potential, to find where tissue neither gains nor loses mass.

TRY IT

Make a solution

You have 1.0 mol dm^{-3} sucrose stock. How would you make 20 cm^3 of 0.3 mol dm^{-3} sucrose solution?

> Hint: The final solution is 0.3 times the stock concentration, so 0.3 of its volume must be stock.

The answer is on the next slide.

NIAID Ryan Kissinger, Public domain

ANSWER

Make a solution

1. Stock: $0.3 \times 20 = 6\ cm^3$
2. Water: $20 - 6 = 14\ cm^3$
3. Measure each with a pipette or syringe, and mix well

Compare your working with each step.

Key point

Check: 6 cm^3 of 1.0 mol dm^{-3} holds the same amount of sucrose as 20 cm^3 of 0.3 mol dm^{-3}.

Chromatography

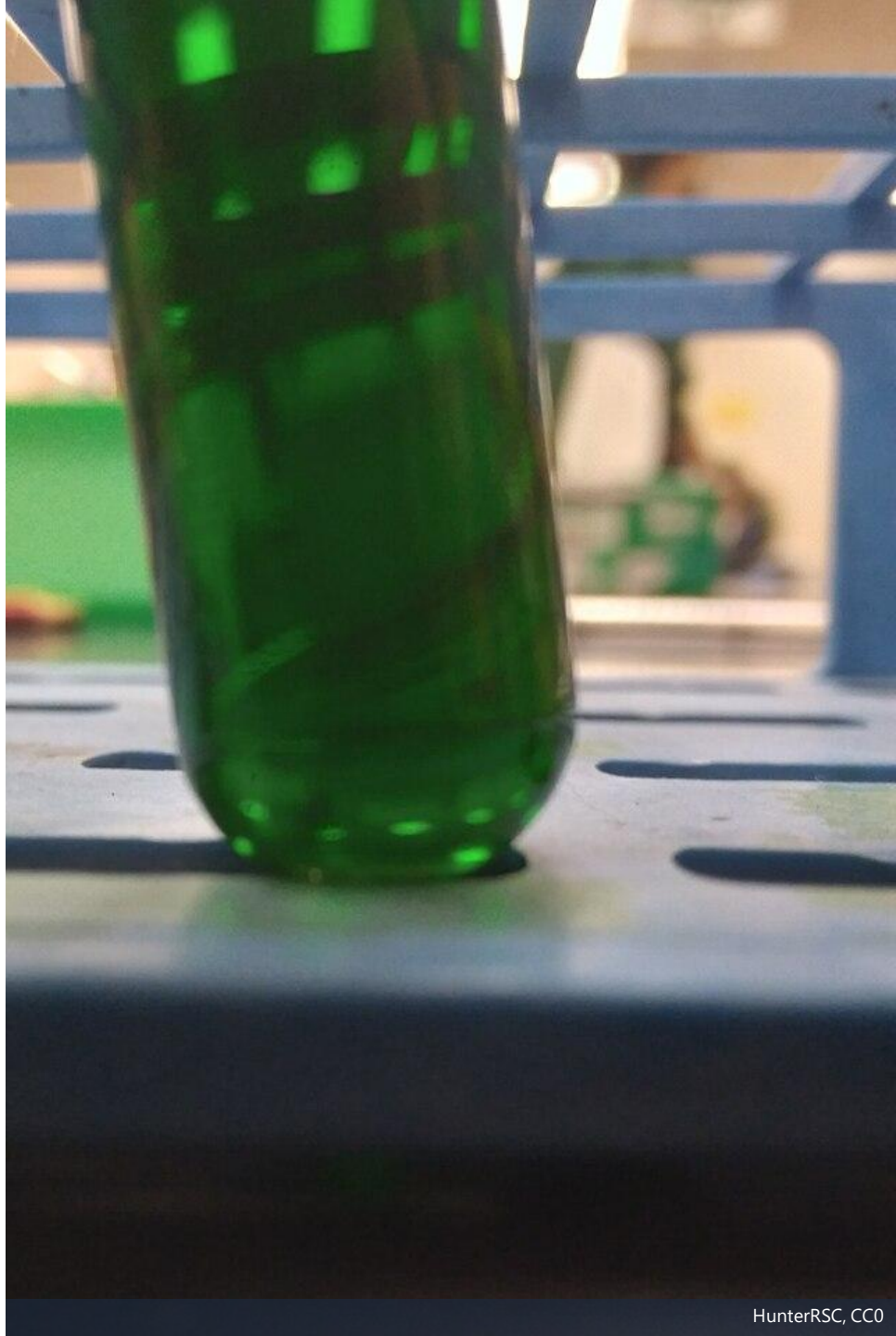
1. Spot the extract on a pencil line
2. Stand in solvent below the line
3. Pigments move at different rates

Rf = spot / solvent front

Distance moved by the spot, divided by the distance moved by the solvent. No units, always below 1.

Worked example

Spot 3.6 cm, solvent front 8.0 cm: Rf = 3.6 / 8.0 = 0.45. Compare with known values to identify pigments. Used in C1.3 Photosynthesis.



HunterRSC, CC0

TRY IT

Calculate Rf

The solvent front moved 9.0 cm. Spot A moved 8.1 cm, spot B 4.5 cm and spot C 2.7 cm.

1. Calculate Rf for each spot.
2. Which pigment is most soluble in the solvent?

The answer is on the next slide.

Oak Ridge National Laboratory, CC BY 2.0

ANSWER

Calculate Rf

Compare your working with each step.

1. A: 8.1 / 9.0 = 0.90
2. B: 4.5 / 9.0 = 0.50
3. C: 2.7 / 9.0 = 0.30
4. A is most soluble in the solvent, so it travels furthest

Key point

Measure to the centre of each spot, and mark the solvent front as soon as you take the paper out.



03

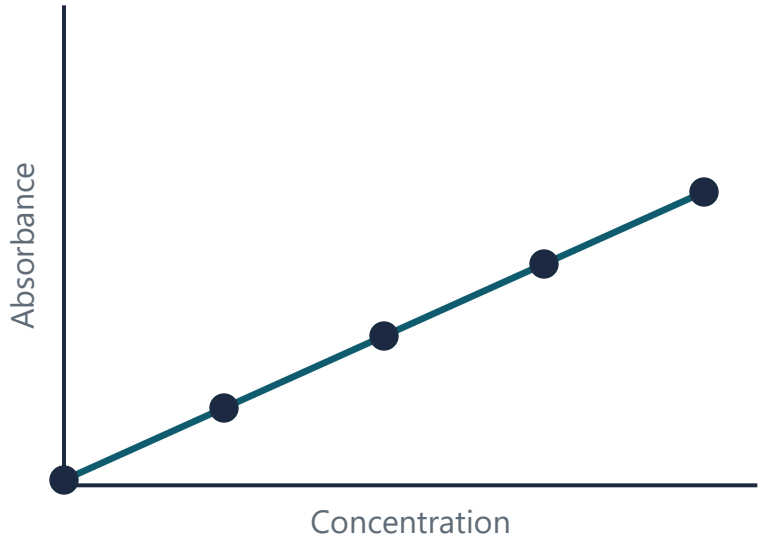
Instruments

Machines that measure colour, gas, and DNA.

U.S. Fish and Wildlife Service – Midwe, Public domain

Colorimetry

- Light of one colour passes through a sample
- The meter reads **absorbance**: more colour absorbs more light
- Zero with a **blank**, then use a **calibration curve**



Used in

C1.1 Enzymes: the starch-iodine colour fades as amylase works.



TimVickers, Public domain

Respirometers and data loggers



Kolathunadu8, CC0

Respirometer

Seeds and a carbon dioxide absorber in a sealed tube. Oxygen use moves a coloured drop (C1.2).



SAHAYA RAJAN S, CC0

pH

Indicator paper gives a rough value. A pH probe and logger give an exact reading every second.


Sébastien Bruneau, Public domain

Indicators

Colour changes show pH, or carbon dioxide in hydrogencarbonate indicator.


QuodScripsiScripsi, CC0

Loggers

Many precise readings, with no missed times. Probes still need calibrating.

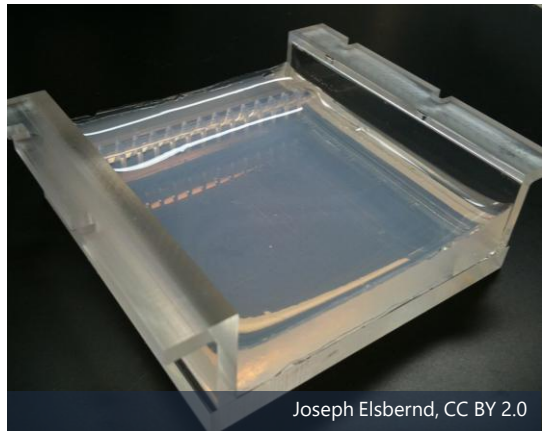
Control

In a respirometer, a second tube with glass beads instead of organisms corrects for changes in temperature and pressure.

PCR and gel electrophoresis

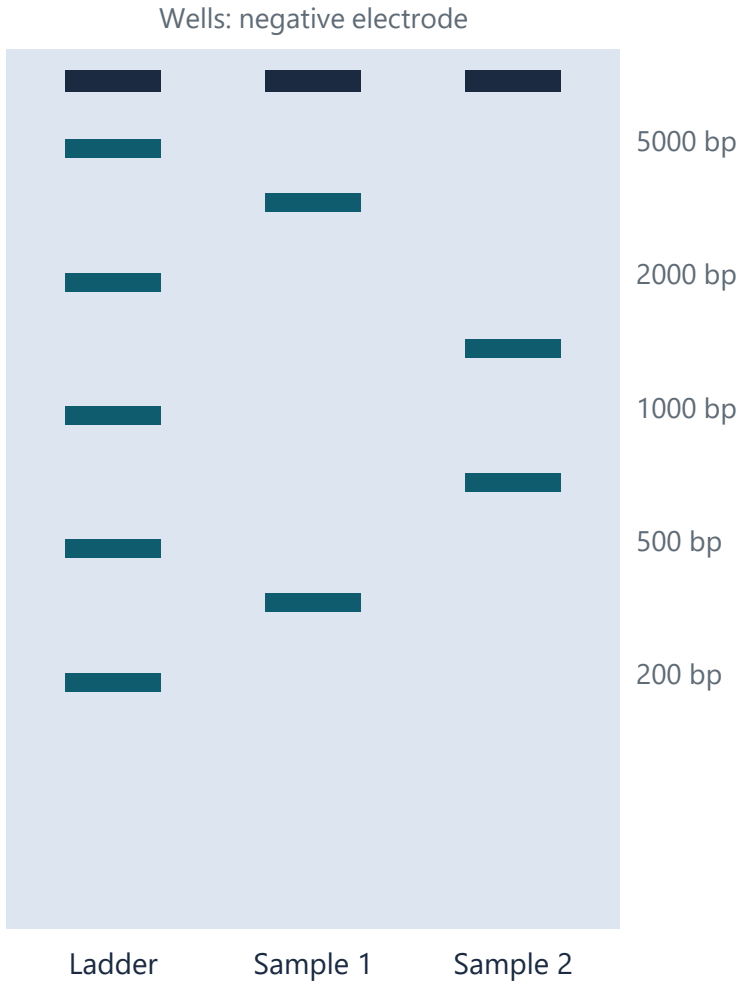


U.S. Fish and Wildlife Service – Midwe, Public domain



Joseph Elsbernd, CC BY 2.0

- **PCR** copies a chosen DNA section by cycles of heating and cooling
- DNA is **negative**, so it moves to the positive electrode
- **Small fragments** move furthest



Read it

Sample 1: bands near 3000 and 300 bp. Sample 2: near 1500 and 700 bp. Compare only with a ladder on the same gel. Used in D1.1.

TRY IT

Read a gel

A ladder has bands at 1000, 800, 600, 400 and 200 bp.

1. A sample band is level with 600 bp. How long is the fragment?
2. Which ends up nearer the wells: 200 or 1000 bp?
3. Why does DNA move through the gel at all?

The answer is on the next slide.

Joseph Elsbernd, CC BY 2.0

ANSWER

Read a gel

1. About 600 base pairs
2. 1000 bp: larger fragments move more slowly
3. DNA has negatively charged phosphate groups, so it moves towards the positive electrode

Compare your working with each step.

Key point

Run a ladder on every gel, because conditions vary between gels.



04

Fieldwork and data

Sample real habitats, then show the results clearly.

Gerda Arendt, CC0

Field techniques



Marie-Noelle De Casamajor, CC BY 4.0

Quadrats

Estimate populations of plants and other sessile organisms (C4.1).



NPS Photo, Public domain

Transects

Sample along a line to see change across a gradient, such as up a shore.



YellowstoneNPS, Public domain

Pitfall traps

Catch ground beetles for capture-mark-release-recapture (C4.1).



DAVID S. FERRY III, CC0

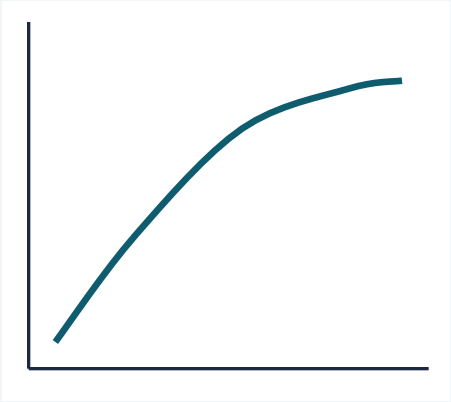
Sweep nets

Sample insects in long grass. Release them afterwards.

Safety

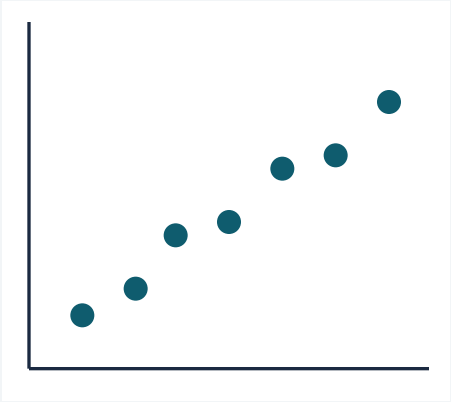
Check tides, weather and footing first. Work in pairs, and wash hands afterwards.

Choosing a graph



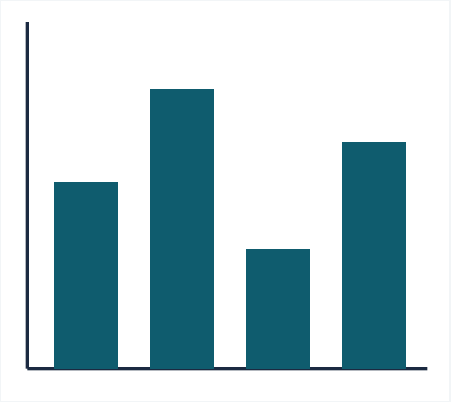
Line graph

Continuous independent variable, showing a trend



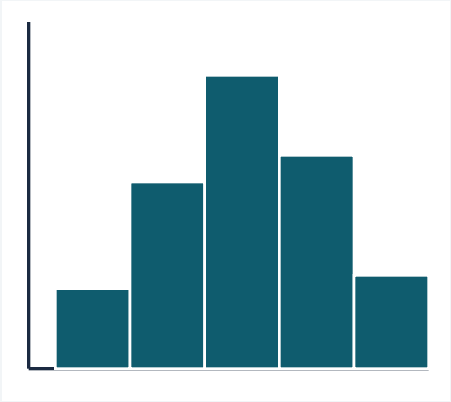
Scatter graph

Correlation between two measured variables



Bar chart

Separate categories, bars apart



Histogram

Frequency in ranges, bars touch

Every graph

Independent variable on the x-axis, labels with units, even scales, and error bars where you have repeats.

TRY IT

Which graph?

Which graph suits each set of data?

1. Mean mass of apples from five varieties.
2. Rate of photosynthesis at light intensities from 0 to 1000 lux.
3. Arm span and height of 30 students.
4. Number of students in each 5 cm band of height.

The answer is on the next slide.

Aethonatic, CC0

ANSWER

Which graph?

1. Bar chart: varieties are categories
2. Line graph: light intensity is continuous, showing a trend
3. Scatter graph: a correlation between two measured variables
4. Histogram: frequency in each range of a continuous variable

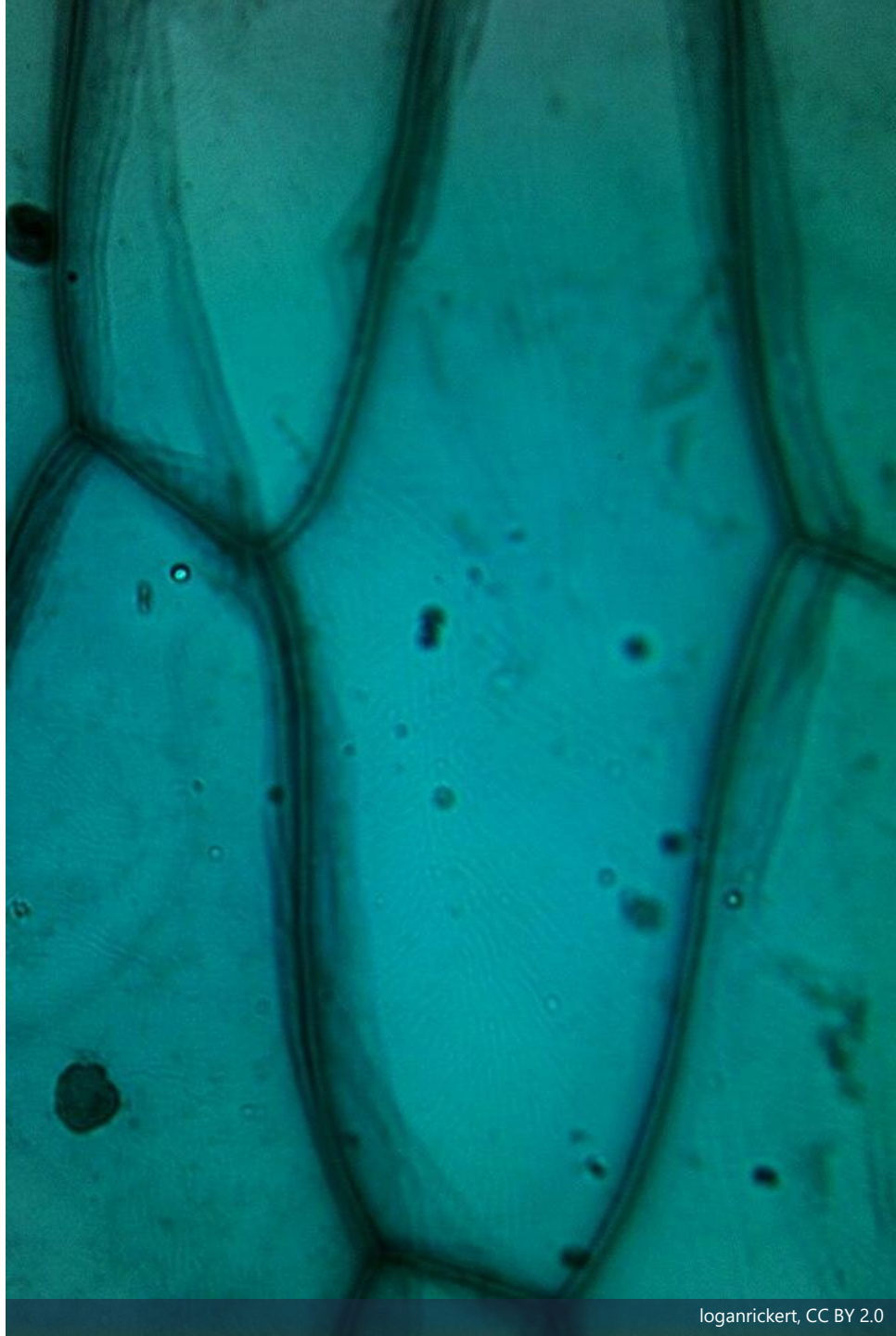
Compare your working with each step.

Key point

In a bar chart the bars are separate. In a histogram they touch, because the ranges are continuous.

Check that you can

1. Prepare and stain a slide, and focus safely
2. Calibrate a graticule and measure a cell
3. State the uncertainty of a reading
4. Make serial and linear dilutions
5. Calculate Rf from a chromatogram
6. Explain colorimeters and respirometers
7. Read a fragment size from a gel
8. Choose a field method and a graph



loganrickert, CC BY 2.0

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. 25 eyepiece units match 5 stage micrometer divisions of 10 µm. What is one eyepiece unit worth?

2. A pigment moves 2.0 cm while the solvent moves 8.0 cm. Calculate its Rf.

3. In gel electrophoresis, which fragments travel furthest, and why?

Image credits

Scientist looking thorugh microscope.jpg – Unknown photographer – Public domain

Microscope (1).jpg – Unknown photographer – Public domain

Test tubes containing chemicals.jpg – Jahra Tasfia Reza – CC0

Whitney Genetics Lab PCR Machine (22495358393).jpg – U.S. Fish and Wildlife Service – Midwest Region – Public domain

Carabus taedatus (top view) (49917028712).jpg – YellowstoneNPS – Public domain

Onion Epidermis Cells W.M. 40x – 249.jpg – loganrickert – CC BY 2.0

Stage Micrometer 01.jpg – RIT RAJARSHI – CC BY 4.0

Calibrated Burette System (08010614) (31702391337).jpg – IAEA Imagebank – CC BY 2.0

Spinach chromatography 20211026 093630.jpg – HunterRSC – CC0

Cuvette holder.jpg – TimVickers – Public domain

Two percent Agarose Gel in Borate Buffer cast in a Gel Tray (Front, angled).jpg – Joseph Elsbernd – CC BY 2.0

Quadrats sur le site Abbadia (Ifremer 00380-49168 – 1950).jpg – Marie-Noelle De Casamajor – CC BY 4.0

Human epithelial cells stained with methylene blue dye.jpg – Daliasalloum – CC BY 4.0

Pipetting activities.jpg – Diane A. Reid (Photographer) – Public domain

Laboratory water bath.jpg – QuodScripsiScripsi – CC0

Stoppuhr hanhart.jpg – Stefan (Diskussion). Original uploader was StefanPohl at Ger – Public domain

Erlenmeyer Flask (Filled) (NIH BioArt 631 – 660919).png – Courtesy of NIAID Ryan Kissinger – Public domain

Thin-layer chromatography (4293629352).jpg – Oak Ridge National Laboratory – CC BY 2.0

Lagerstroemia microcarpa seed germinating 10.jpg – Kolathunadu8 – CC0

Ph test strips.jpg – SAHAYA RAJAN S – CC0

Indicateurs colorés de pH.jpg – Sébastien Bruneau – Public domain

Meadow of wildflowers, Oberauroff.jpg – Gerda Arendt – CC0

Ornithologist studies birds in Glacier Bay (2fcdf2ed-0388-4be7-9da8-1b7c0d4af8a7).jpg – NPS Photo – Public domain

Ladybug eating aphids. (49485764276).jpg – DAVID S. FERRY III from PLANT CITY,FL., U.S.A. – CC0

Pile of leaves – 11 October 2026.jpg – Aethonatic – CC0

All from Wikimedia Commons (commons.wikimedia.org, search the file name). Some images are cropped. CC licences: creativecommons.org/licenses